

Reminders

Upcoming due dates

Fri Discussion lab 3

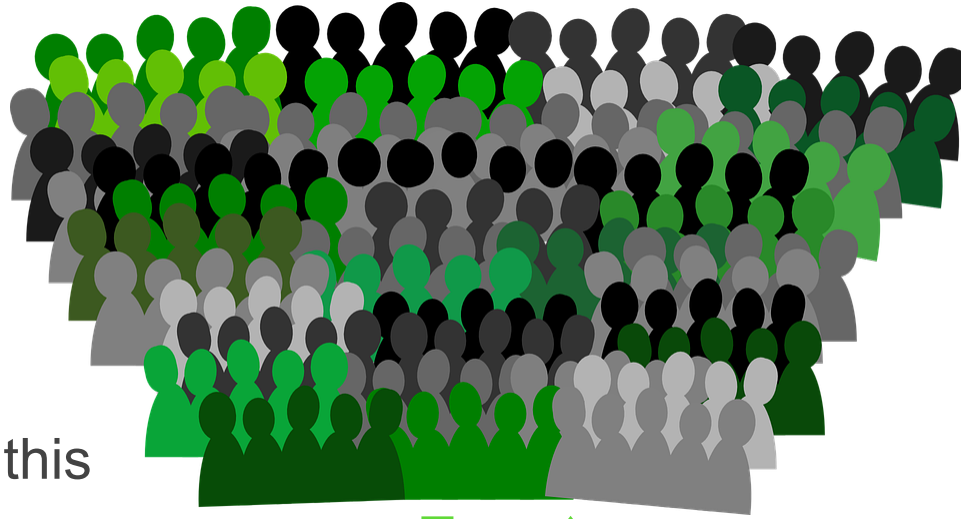
Repo invites! Click accept before it expires!

Section this week covers data visualization

Statistical Inference 1

Data Science in Practice

Population



We want to learn something about this

Sampling



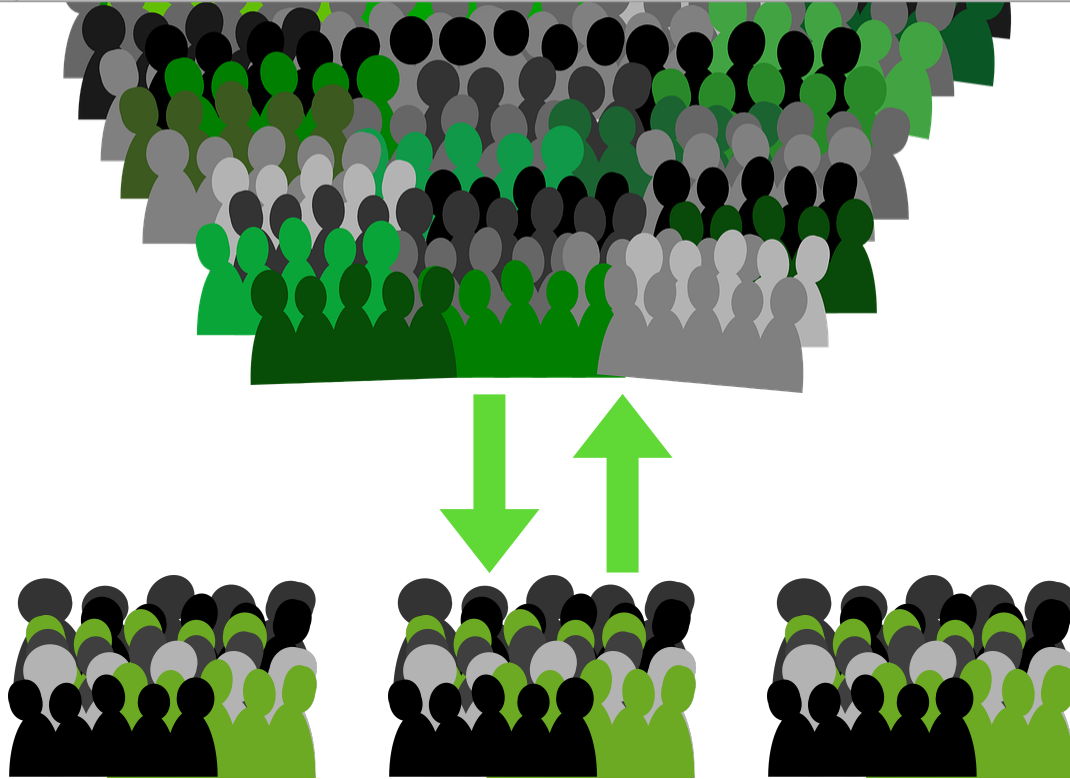
Inference

....but we can only *actually* collect data from this



Sample

Random samples are random!
They differ from each other and the population!





Published in final edited form as:

Epidemiology. 2013 January ; 24(1): 23–31. doi:10.1097/EDE.0b013e3182770237.

The Effect of Air Pollution Control on Life Expectancy in the United States: An Analysis of 545 US counties for the period 2000 to 2007

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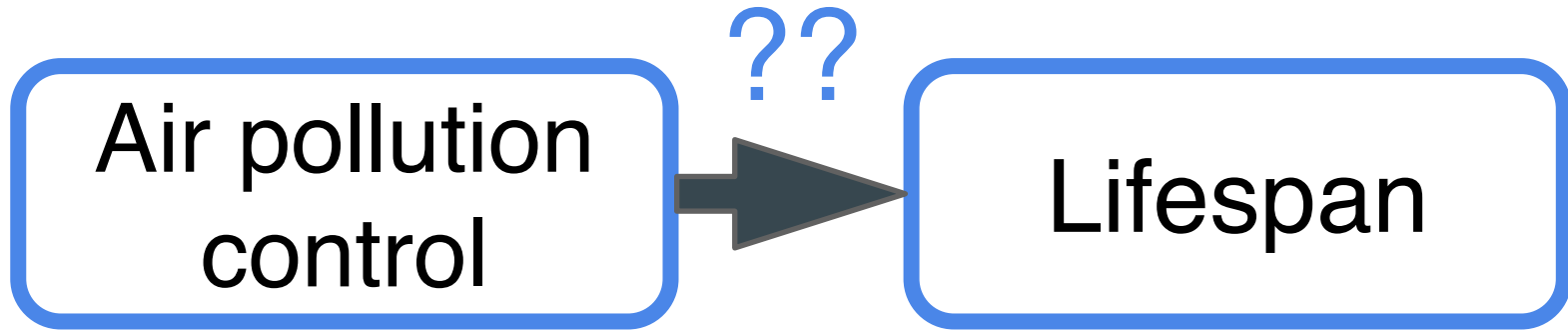
Department of Biostatistics, Harvard School of Public Health, 655 Huntington Avenue, HSPH Building 2, 4th Floor, Boston, MA 02115

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Is there a relationship between air pollution control and lifespan?

A decrease of 10 $\mu\text{g}/\text{m}^3$ in the concentration of $\text{PM}_{2.5}$ was associated with an increase in mean life expectancy of 0.35 years (SD= 0.16 years, $p = 0.033$). This association was stronger in more urban and densely populated counties.

What if we want to know the effect of air pollution on everyone in the United States?



The population:
every individual in the USA

The sample:
545 US counties



#	State	Total Number of Counties	Total Area
1	Texas	254	268,596 mi ²
2	Georgia	159	59,425 mi ²
3	Virginia	133	42,775 mi ²
4	Kentucky	120	40,408 mi ²
5	Missouri	115	69,707 mi ²
6	Kansas	105	82,278 mi ²
7	Illinois	102	57,914 mi ²
8	North Carolina	100	53,819 mi ²
9	Iowa	99	56,273 mi ²
10	Tennessee	95	42,144 mi ²

3143
counties
in the US

CA is #27

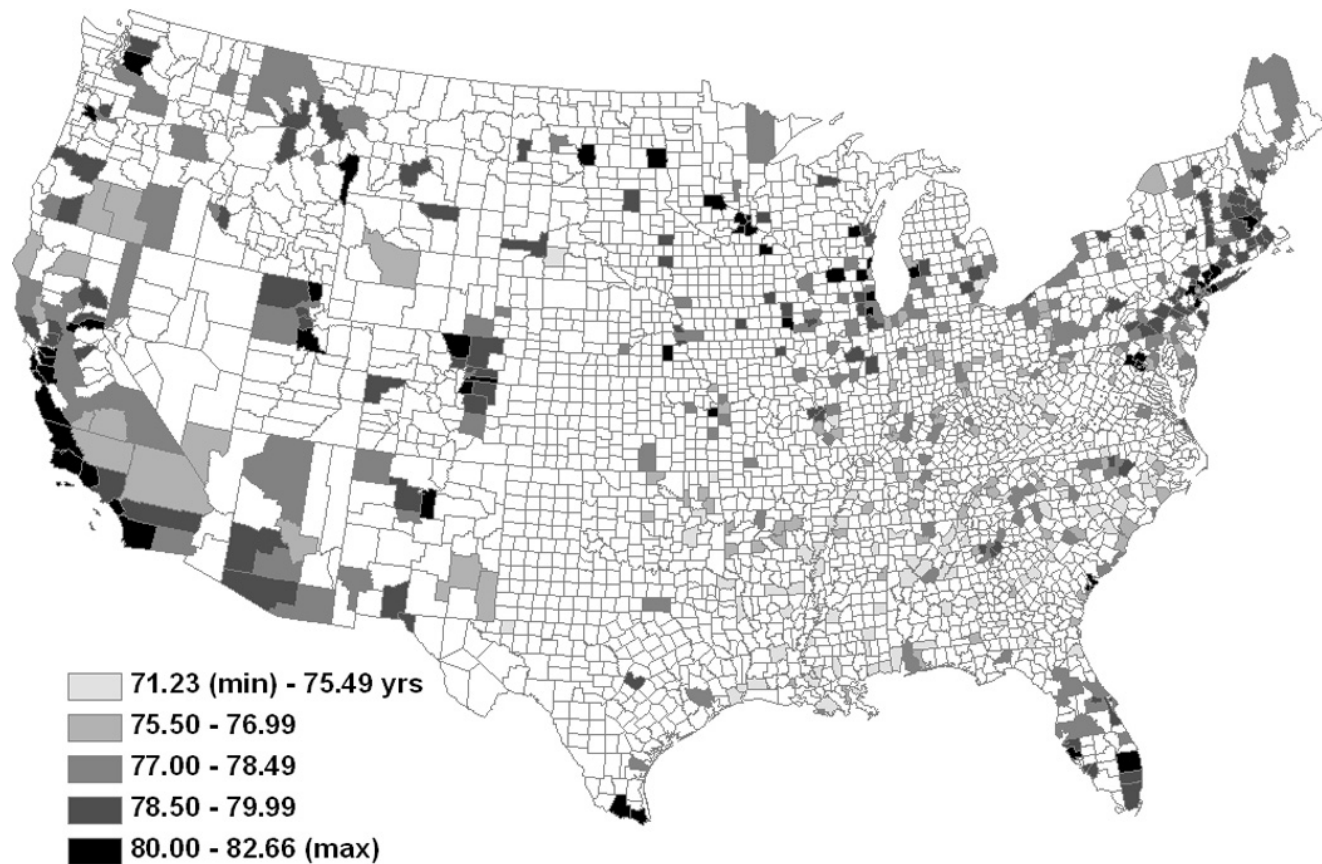
58 counties on 163k mi²

Random Sampler

How would you want to select those 545 counties?

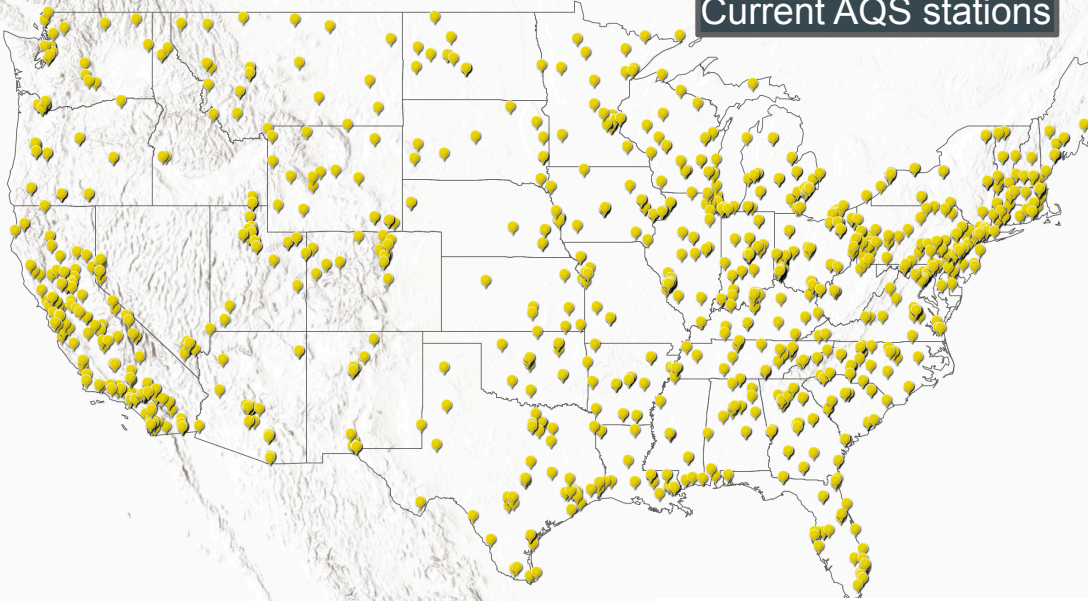
What criteria are ideal?

What do you think they did?



All counties with with available matching PM2.5 data for 2000 and 2007 from the EPA's Air Quality System. Includes both metropolitan and non-metro counties

Current AQS stations



States typically decide where monitors are placed based on areas of relatively high population and/or areas believed to have relatively higher pollutant concentrations. Each state is responsible for developing its own monitoring plan, which is then reviewed and revised every five years. Aug 28, 2023



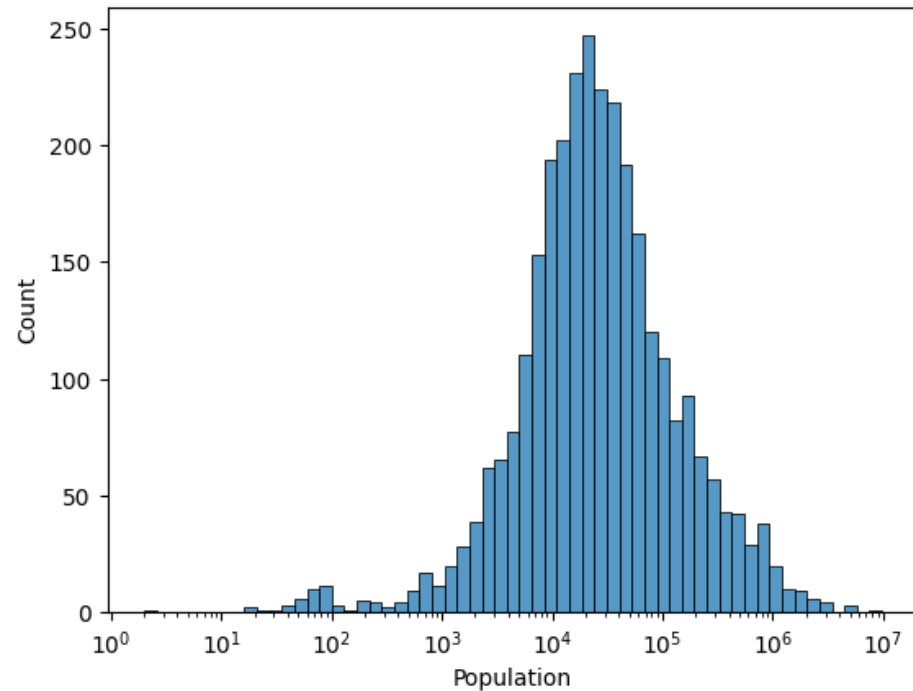
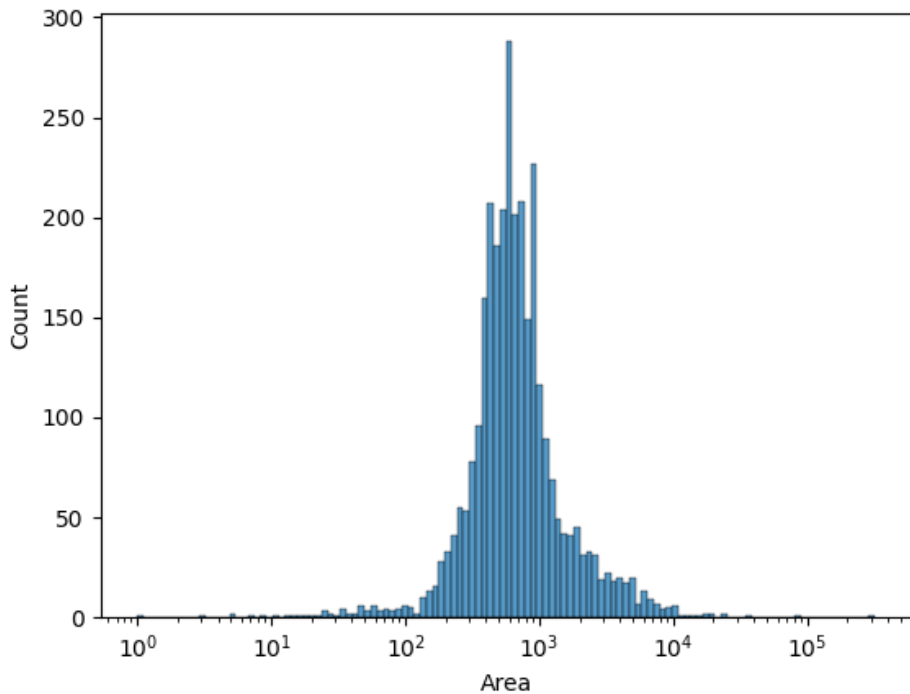
United States Environmental Protection Agency (.gov)

<https://www.epa.gov/outdoor-air-quality-data/who-decides-where-monitors-get-placed>



Who decides where monitors get placed? | US EPA

Distributions of land area and population size among US counties



Random Sampler

How might conclusions differ if...

the sample locations were instead completely spatially at random across the USA?

The sample came from the 1980s or the 2020s instead of the 2000s?

Approaches to Inference

CORRELATION

COMPARISON OF MEANS

REGRESSION

NON-PARAMETRIC TESTS

CORRELATION

ASSOCIATION
BETWEEN VARIABLES

i.e. Pearson
Correlation,
Spearman
Correlation, chi-
square test

COMPARISON OF MEANS

DIFFERENCE IN MEANS
BETWEEN VARIABLES

i.e. t-test, ANOVA

REGRESSION

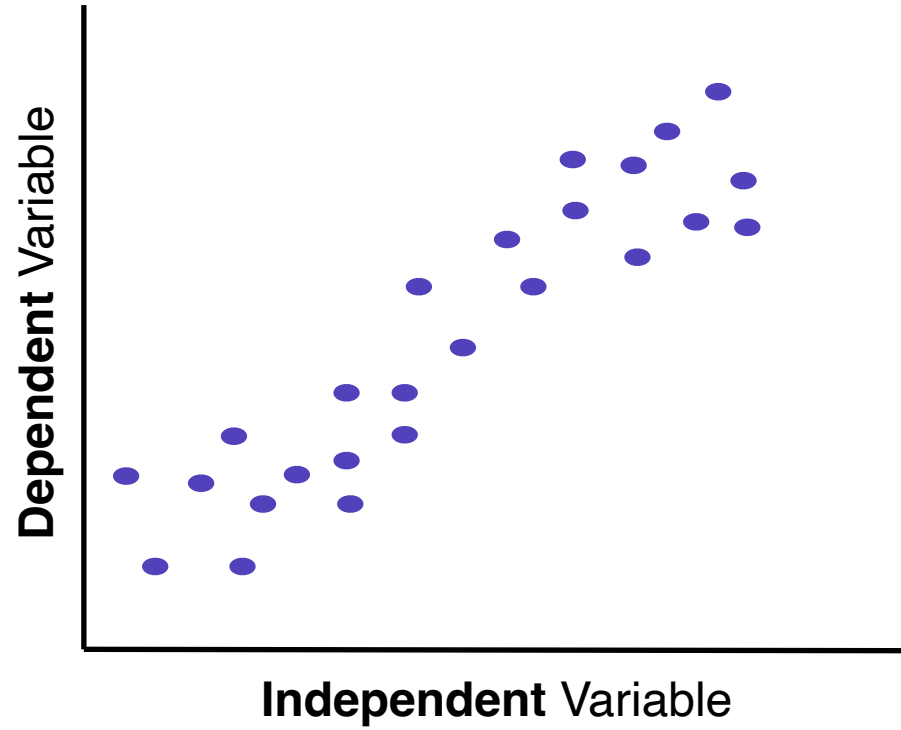
DOES CHANGE IN ONE
VARIABLE MEAN CHANGE
IN ANOTHER?

i.e. simple
regression, multiple
regression

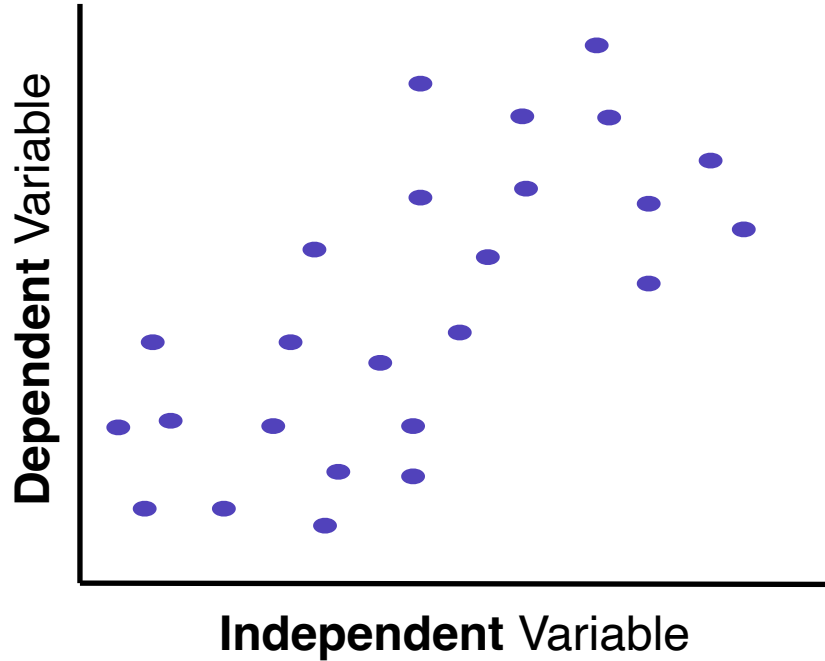
NON-PARAMETRIC TESTS

FOR WHEN ASSUMPTIONS
IN THESE OTHER 3
CATEGORIES ARE NOT
MET

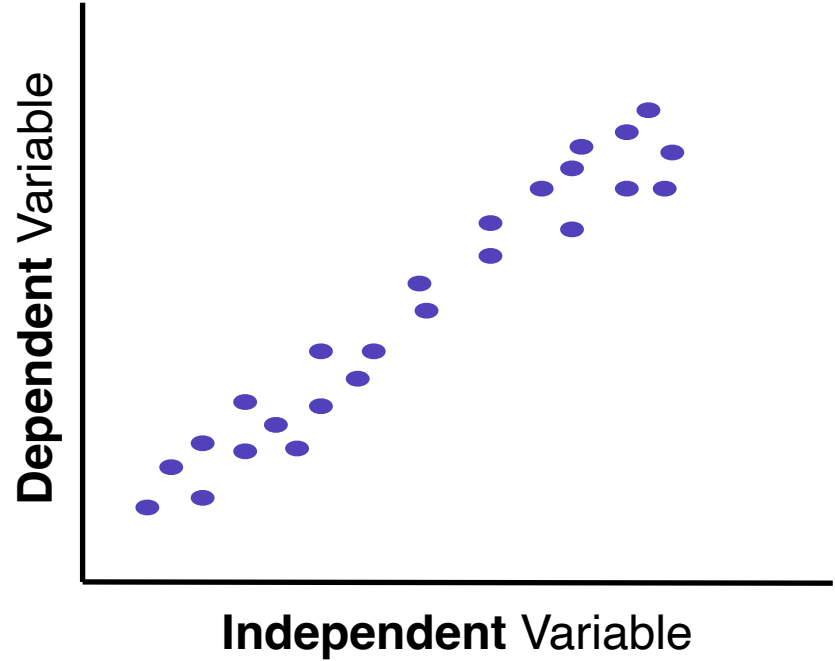
i.e. Wilcoxon rank-
sum test, Wilcoxon
sign-rank test, sign
test



weaker relationship

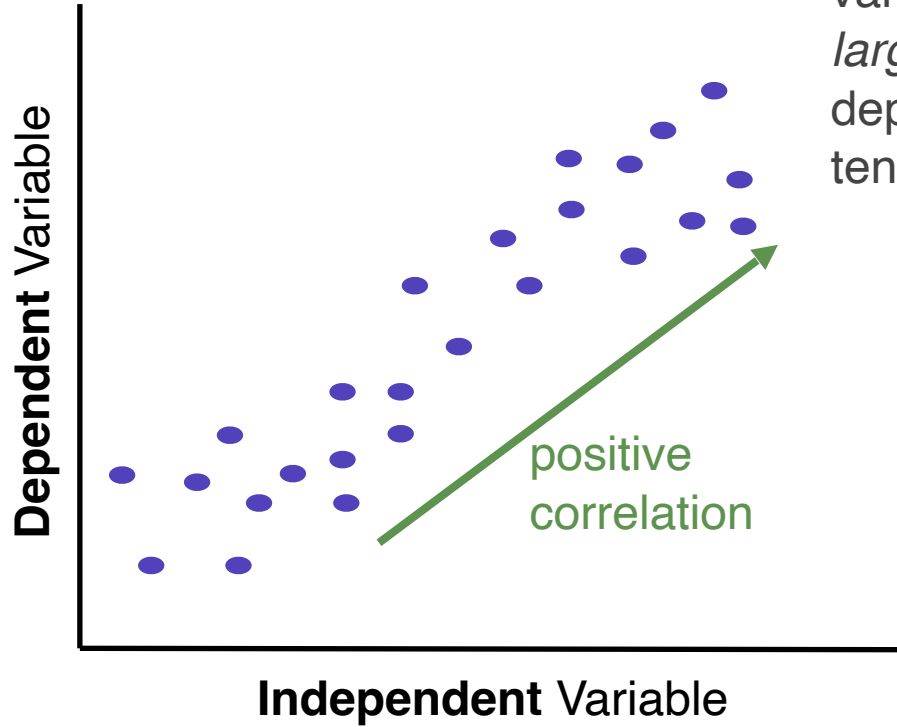


stronger relationship



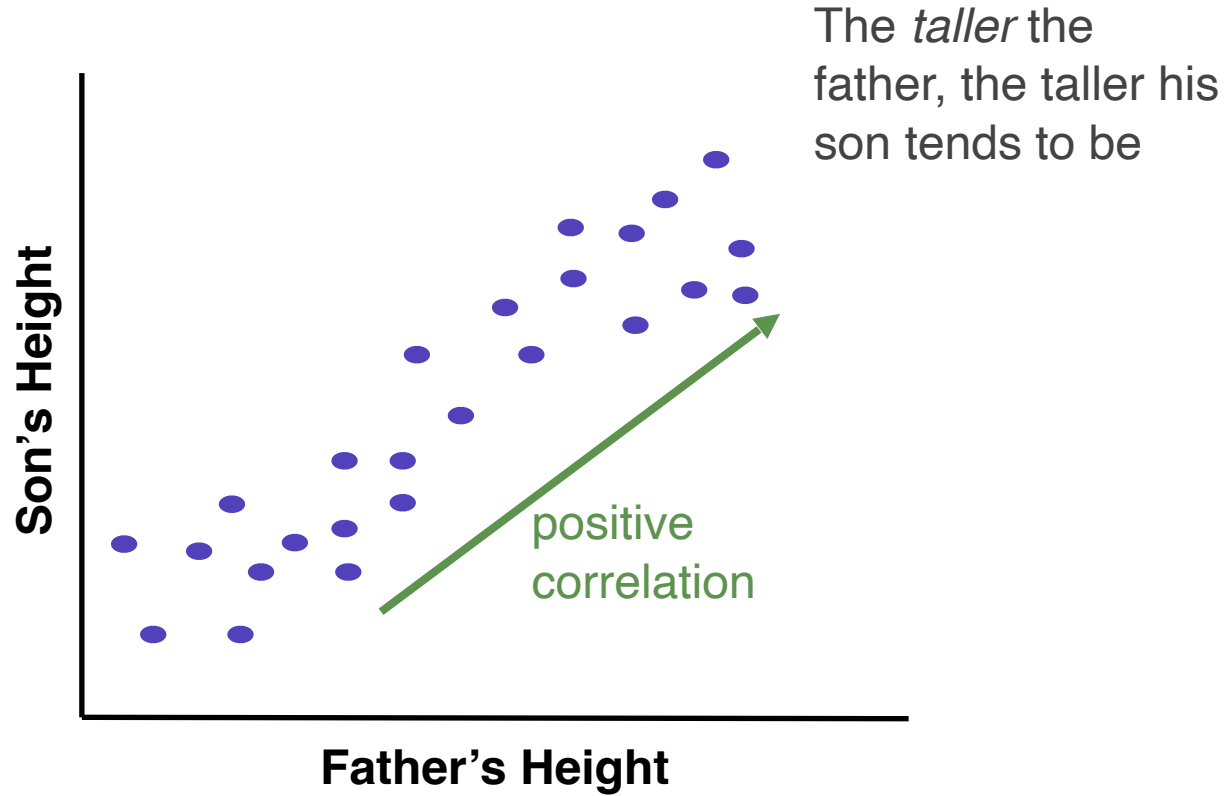
stronger relationship = higher correlation

The *smaller* the independent variable value, the *smaller* the dependent variable tends to be

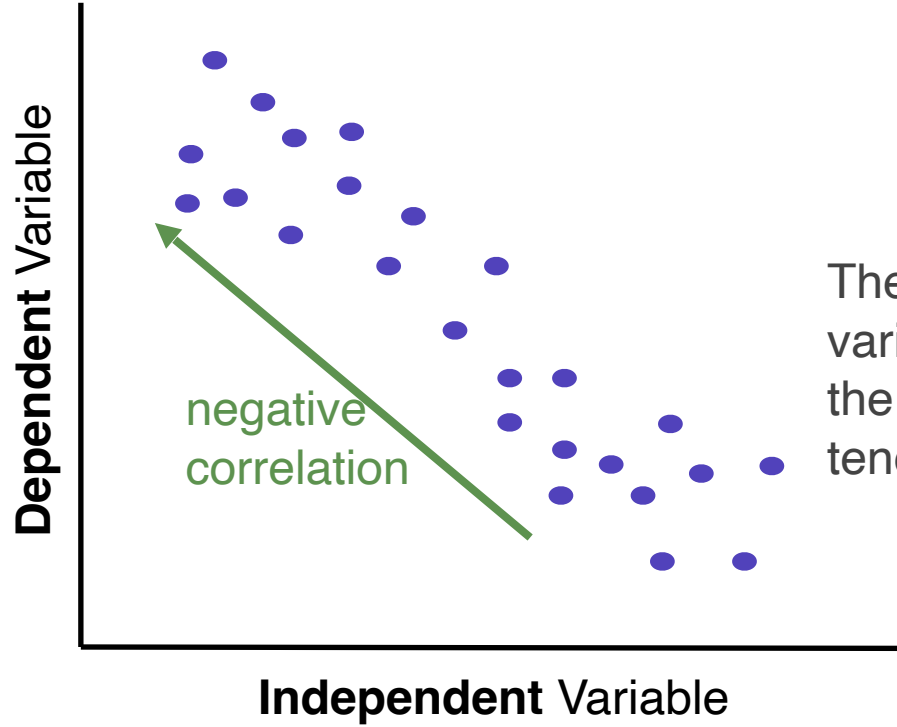


The *larger* the independent variable value, the *larger* the dependent variable tends to be

The *shorter* the father, the shorter his son tends to be

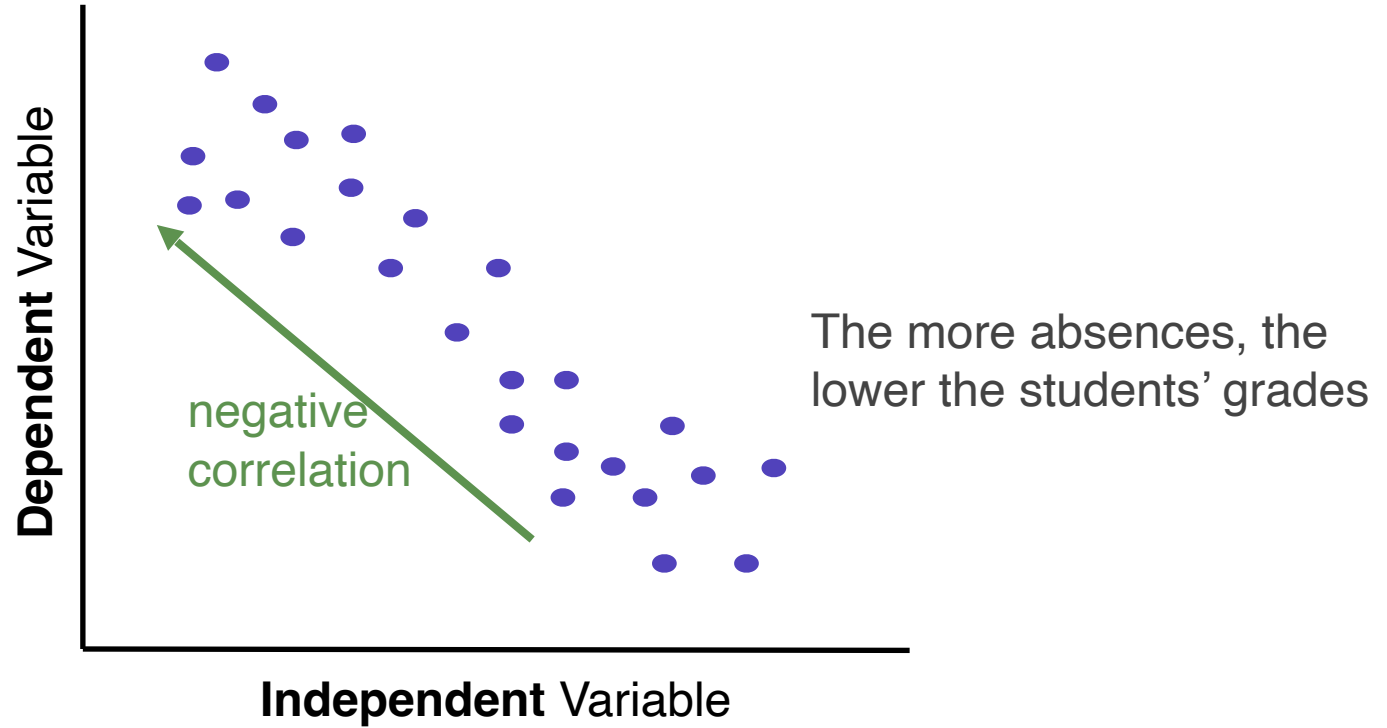


The *smaller* the independent variable value, the *larger* the dependent variable tends to be



The *larger* the independent variable value, the *smaller* the dependent variable tends to be

The *lower* the number of absences, the *higher* the students' grades tend to be



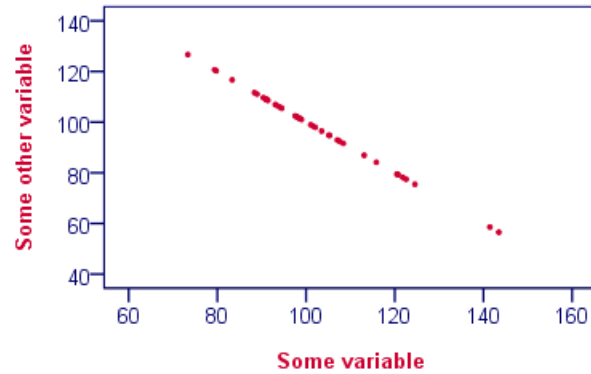
Pearson's r :

linear correlation between two variables

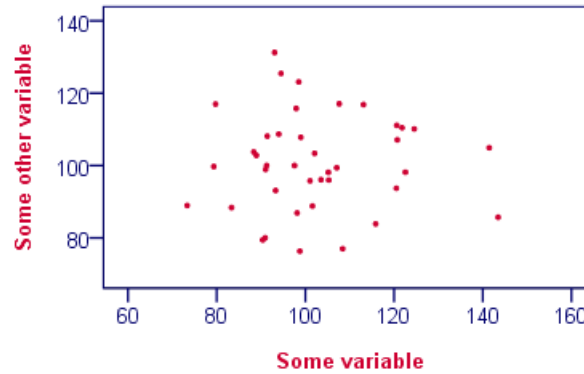
takes values $[-1, 1]$

Correlation is how close the data are to being in a line...
BUT IT HAS NOTHING TO DO WITH THE SLOPE

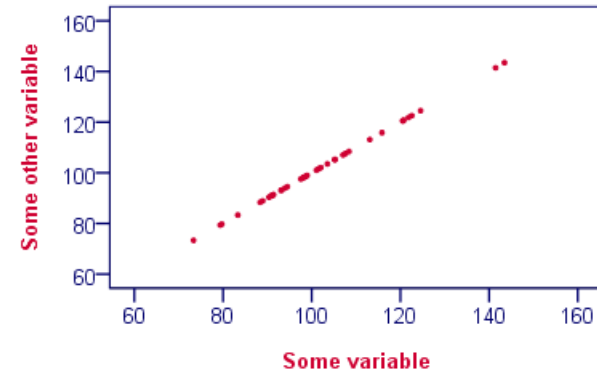
Correlation Coefficient = -1

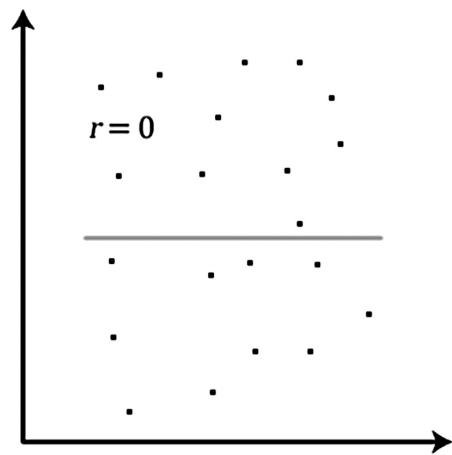
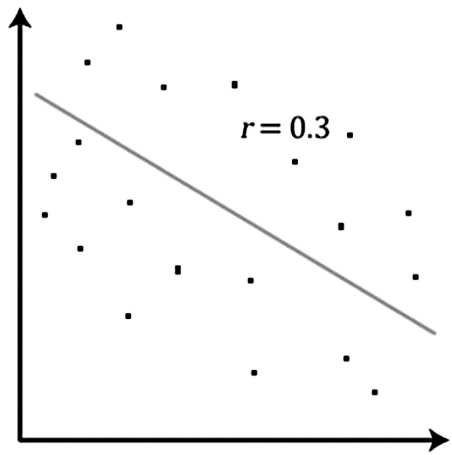
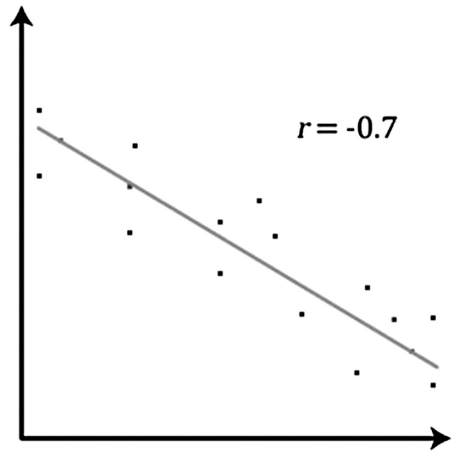
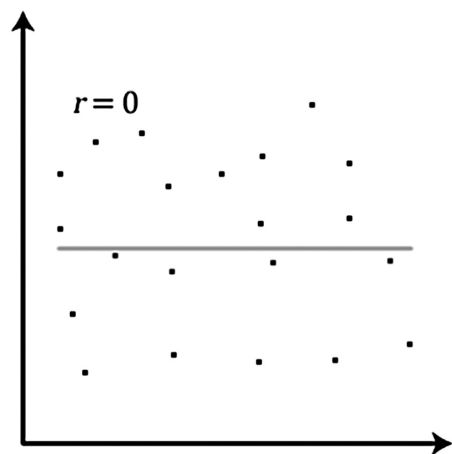
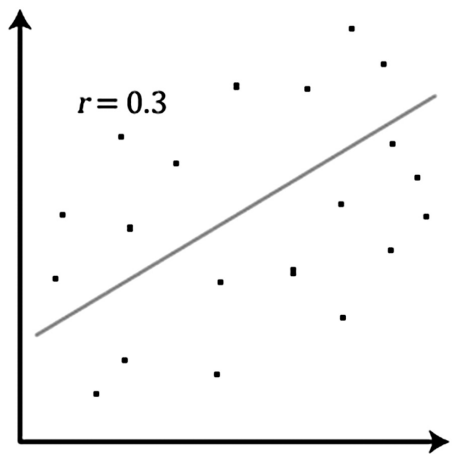
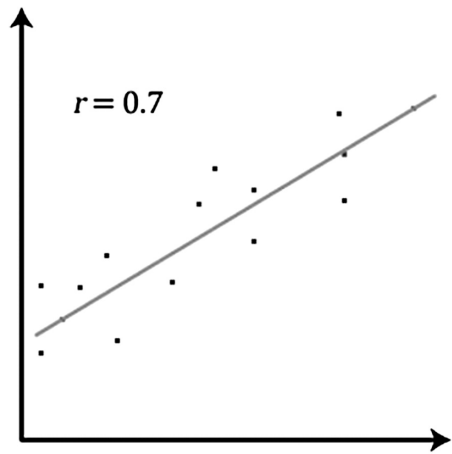


Correlation Coefficient = 0



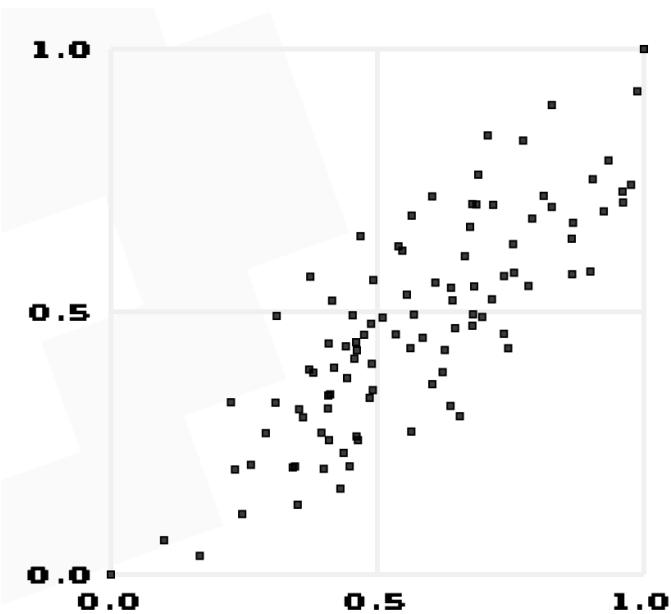
Correlation Coefficient = 1





<https://www.guessthecorrelation.com/>

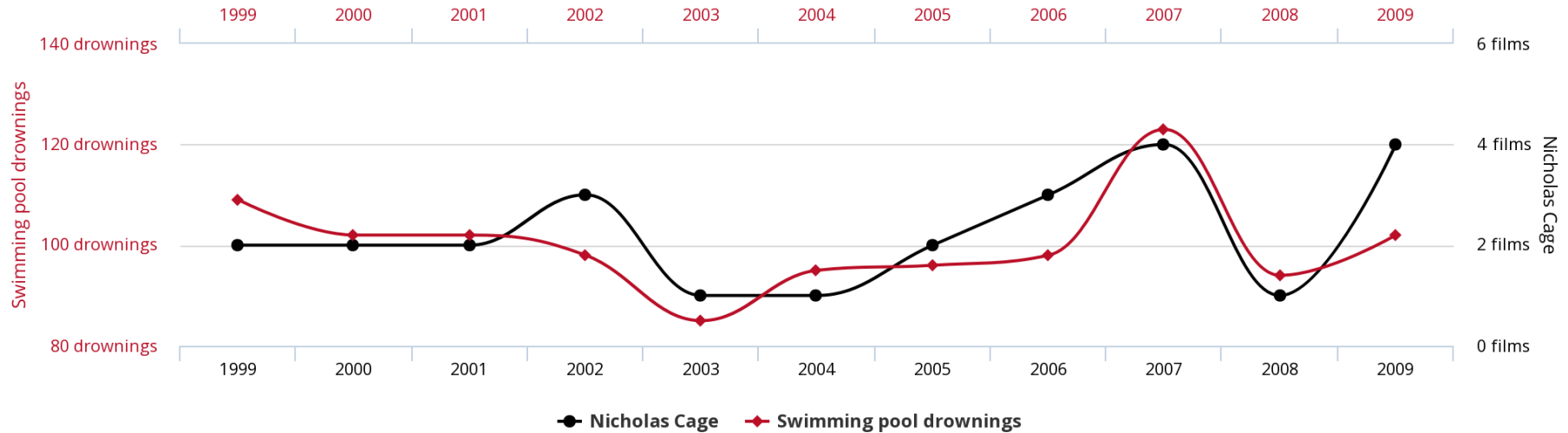
Which of the following is the Pearson correlation coefficient (r) for this relationship?



<https://forms.gle/M4Q3hFkeoLKCixUX8>

Correlation != Causation

Number of people who drowned by falling into a pool
correlates with
Films Nicolas Cage appeared in



tylervigen.com

What about correlation between categorical variables?

- Make a contingency table ...

Sex \ Handed-ness	Right-handed	Left-handed	Total
Male	43	9	52
Female	44	4	48
Total	87	13	100

- Then calculate phi (based on chi-squared statistic) ...

Phi coefficient [\[edit \]](#)

Main article: [Phi coefficient](#)

A simple measure, applicable only to the case of 2×2 contingency tables, is the [phi coefficient](#) (ϕ) defined by

$$\phi = \pm \sqrt{\frac{\chi^2}{N}},$$

where χ^2 is computed as in [Pearson's chi-squared test](#), and N is the grand total of observations. ϕ varies from 0 (corresponding to no association between the variables) to 1 or -1 (complete association or complete inverse association), provided it is based on frequency data represented in 2×2 tables. Then its sign equals the sign of the product of the [main diagonal](#) elements of the table minus the product of the off-diagonal elements. ϕ takes on the minimum value -1.0 or the maximum value of $+1.0$ [if and only if](#) every marginal proportion is equal to 0.5 (and two diagonal cells are empty).^[2]

- Yes there are ways to do this in pandas/numpy and other libraries!

CORRELATION

ASSOCIATION
BETWEEN VARIABLES

i.e. Pearson
Correlation,
Spearman
Correlation, chi-
square test

COMPARISON OF MEANS

DIFFERENCE IN MEANS
BETWEEN CONDITIONS

i.e. t-test, ANOVA

REGRESSION

DOES CHANGE IN ONE
VARIABLE MEAN CHANGE
IN ANOTHER?

I.e. simple
regression, multiple
regression

NON-PARAMETRIC TESTS

FOR WHEN ASSUMPTIONS
IN THESE OTHER 3
CATEGORIES ARE NOT
MET

i.e. Wilcoxon rank-
sum test, Wilcoxon
sign-rank test, sign
test

t-test:

tests for difference in means between groups



William Sealy Gosset (13 June 1876 – 16 October 1937) was an English statistician, chemist and brewer who served as [Head Brewer of Guinness](#) and Head Experimental Brewer of Guinness and was a pioneer of modern statistics. He pioneered small sample experimental design and analysis with an economic approach to the logic of uncertainty. Gosset published under the [pen name Student](#) and developed most famously [Student's t-distribution](#) – originally called Student's "z" – and "Student's test of [statistical significance](#)".^[1]

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Life and career [[edit](#)]

Born in [Canterbury](#), England the eldest son of Agnes Sealy Vidal and Colonel Frederic Gosset, R.E. [Royal Engineers](#), Gosset attended [Winchester College](#) before matriculating as Winchester Scholar in [natural sciences](#) and mathematics at [New College, Oxford](#). Upon graduating in 1899, he joined the brewery of [Arthur Guinness](#) & Son in [Dublin](#), Ireland; he spent the rest of his 38-year career at Guinness.^{[1][2]}

Gosset had three children with [Marjory Gosset](#) (née Phillpotts). Harry Gosset (1907–1965) was a consultant paediatrician; Bertha Marian Gosset (1909–2004) was a geographer and nurse; the youngest, Ruth Gosset (1911–1953) married the Oxford mathematician Douglas Roaf and had five children.

In his job as Head Experimental Brewer at [Guinness](#), the self-trained Gosset developed new statistical methods – both in the brewery and on the farm – now central to the design of experiments, to proper use of significance testing on repeated trials, and to analysis of **economic significance** (an early instance of [decision theory](#) interpretation of statistics) and more, such as his small-sample, stratified, and repeated balanced experiments on [barley](#) for proving the best [yielding](#) varieties.^[3] Gosset acquired that knowledge by study, by trial and error, by cooperating with others, and by spending two terms in 1906–1907 in the Biometrics laboratory of [Karl Pearson](#).^[4] Gosset and Pearson had a good relationship.^[4] Pearson helped Gosset with the mathematics of his papers, including the 1908 papers, but had little appreciation of their importance. The papers addressed the brewer's concern with small samples; biometricians like Pearson, on the other hand, typically had hundreds of observations and saw no urgency in developing small-sample methods.^[2]

Gosset's first publication came in 1907, "On the Error of Counting with a [Haemocytometer](#)," in which – unbeknownst to Gosset aka "Student" – he rediscovered the [Poisson distribution](#).^[3] Another researcher at Guinness had previously published a paper containing trade secrets of the Guinness

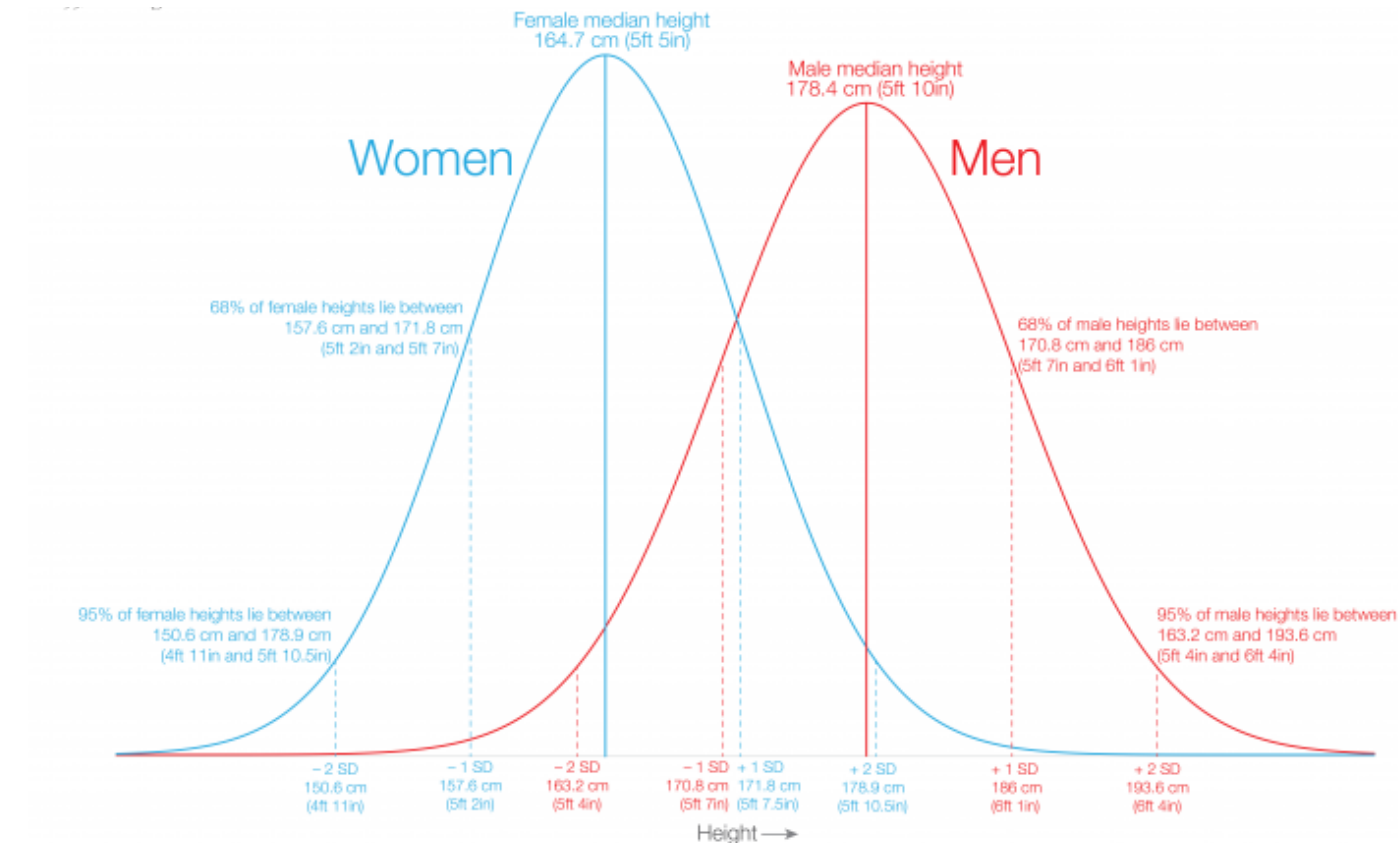
William Sealy Gosset



William Sealy Gosset (aka *Student*) in 1908 (age 32)

Born	13 June 1876 Canterbury, Kent, England
Died	16 October 1937 (aged 61) Beaconsfield, Buckinghamshire, England
Other names	Student
Alma mater	New College, Oxford , Winchester College
Known for	Student's t-distribution , statistical significance , design of experiments , Monte Carlo method , quality control , Modern synthesis , agricultural economics , econometrics
Children	5, including Isaac Henry Gosset
Scientific career	

Do the heights between males and females differ?



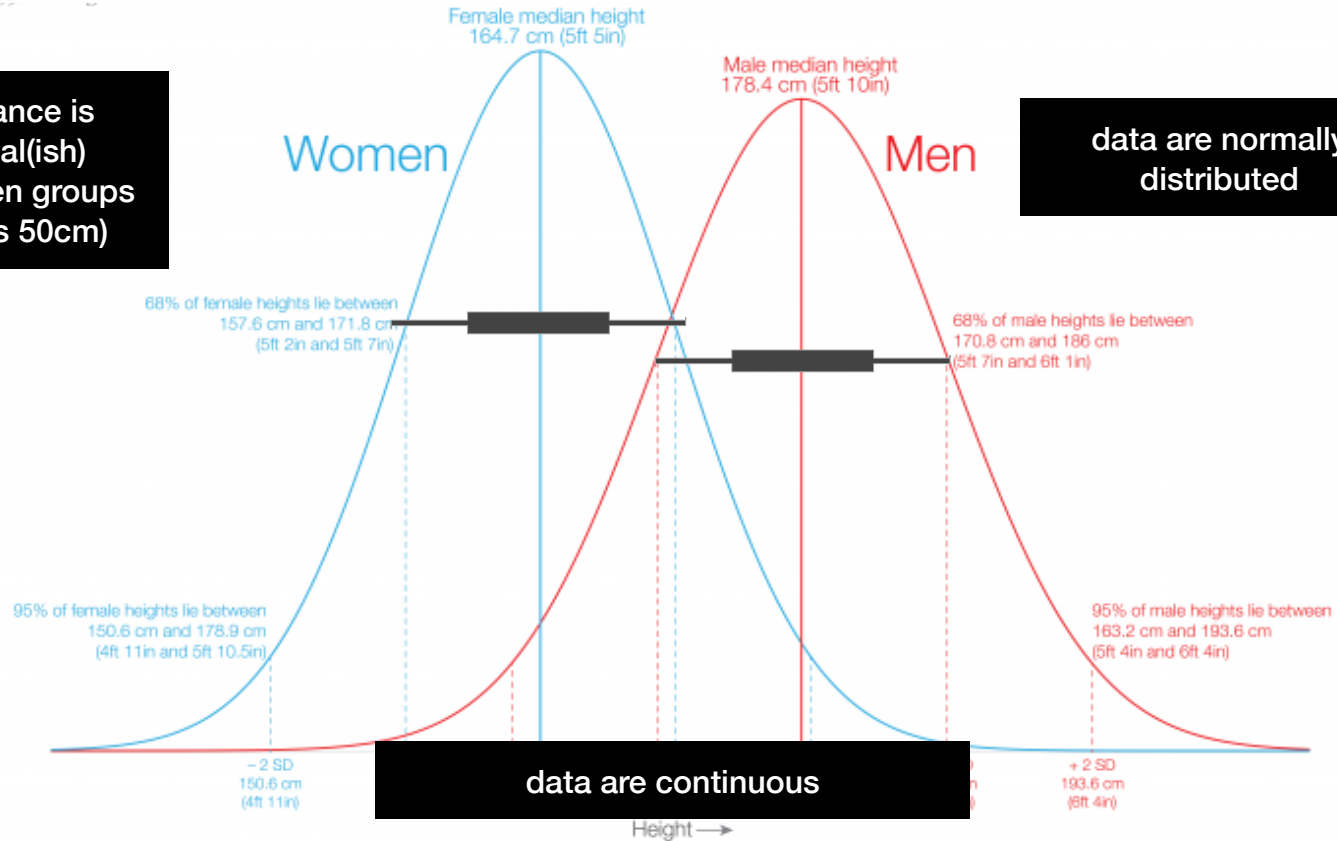
t-test Assumptions

1. Data are continuous
2. Normally distributed
3. Equal variance b/w groups (but can use Welch's test!)
4. Not paired (will talk more about this later)

Do the heights between males and females differ?

variance is
equal(ish)
between groups
(57 vs 50cm)

data are normally
distributed



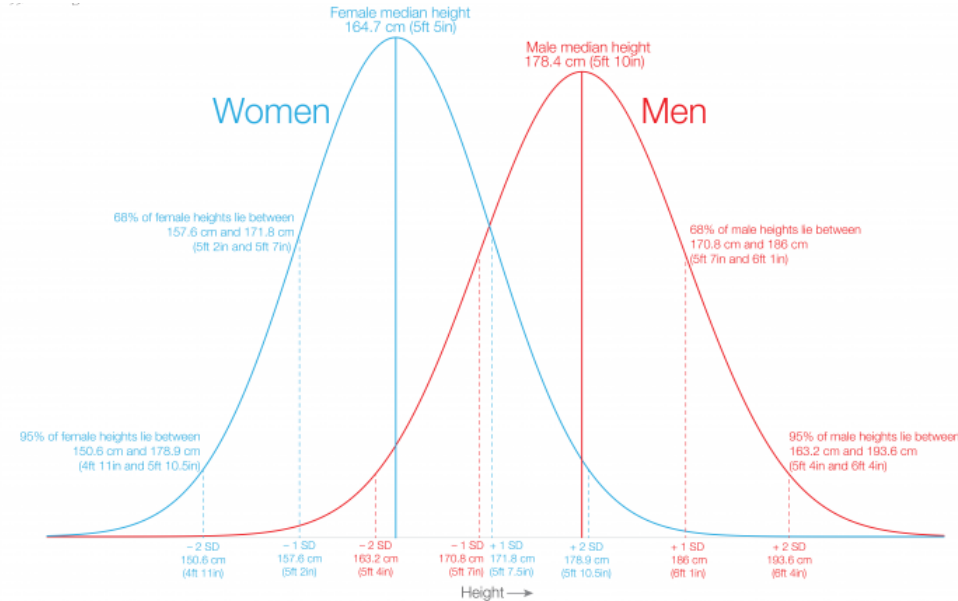
Do the heights between males and females differ?

t-statistic: -95.6

p-value $\ll 0.001$

95% CI for true difference in means
[-5.43, -5.21]

Yes.

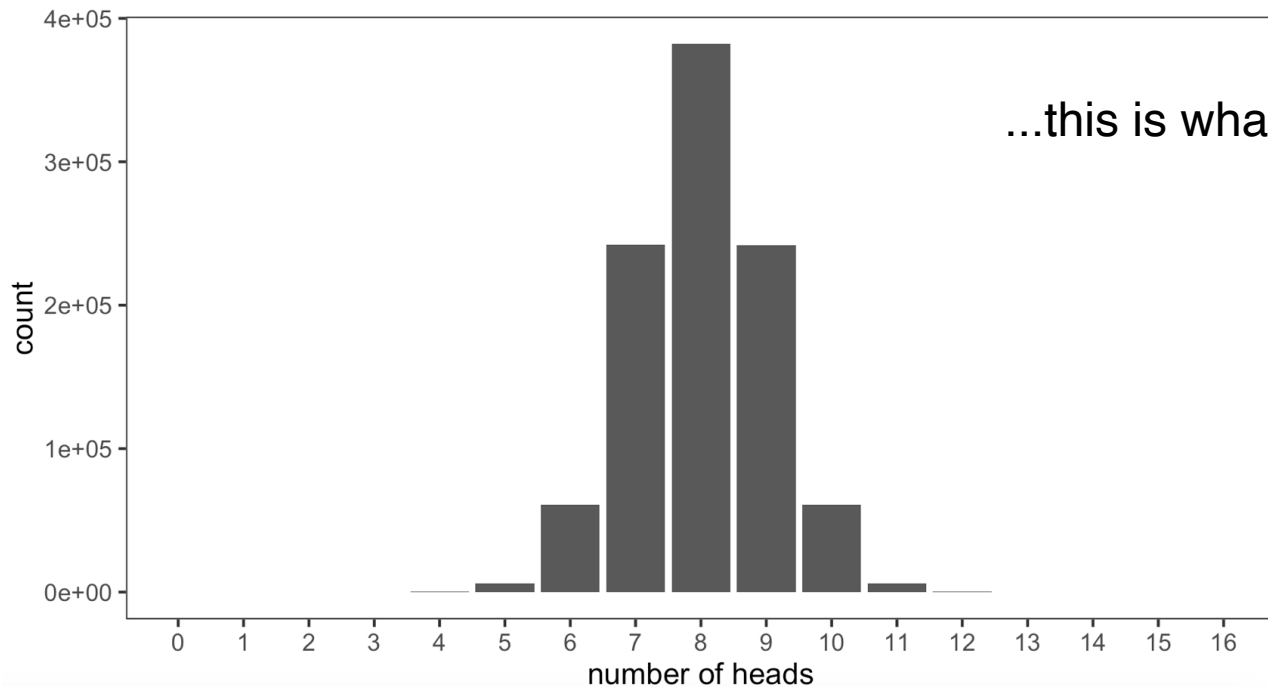


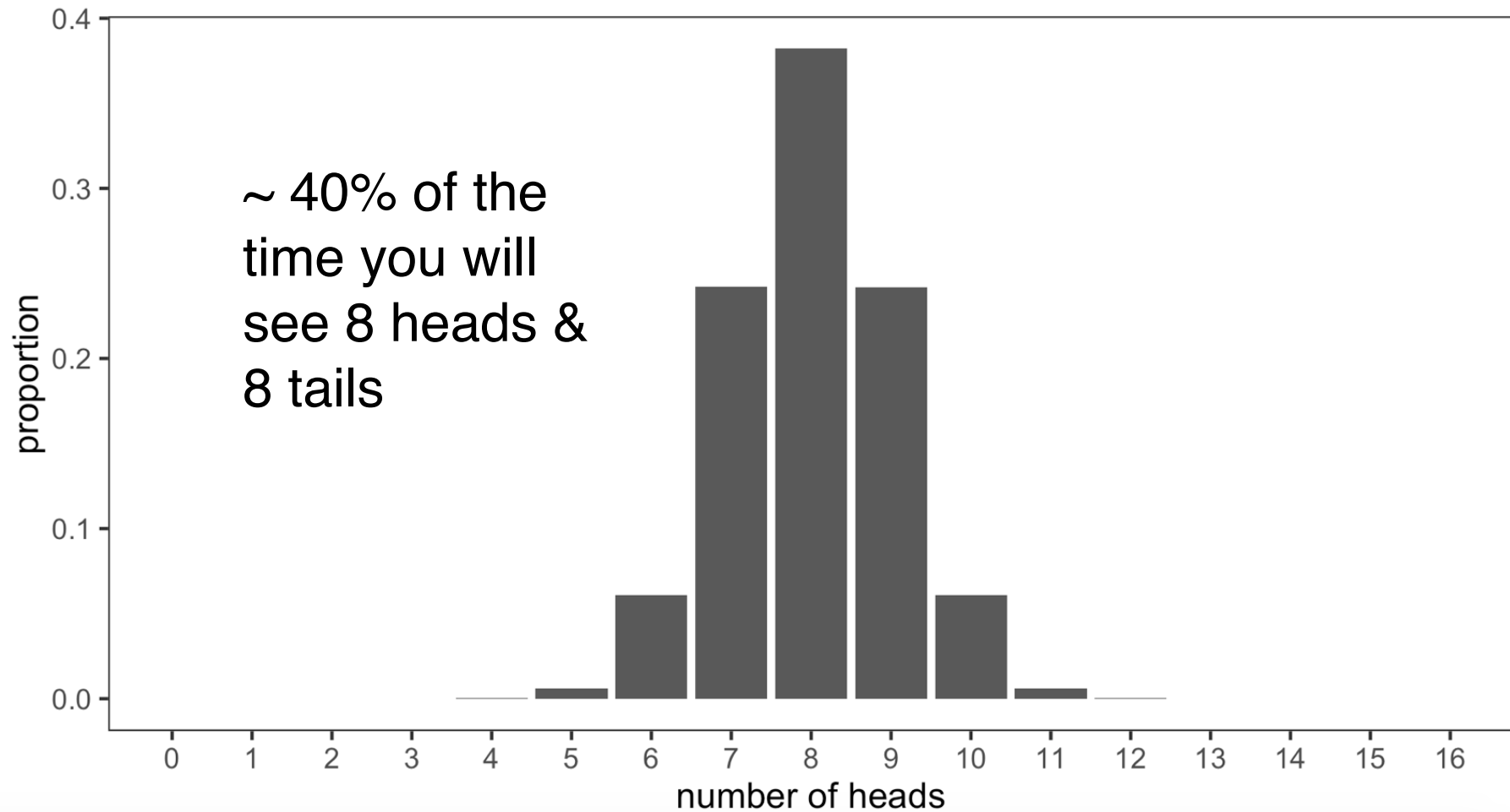
p-value : the probability under the **null hypothesis** of getting measurements as extreme as the observed results by chance alone

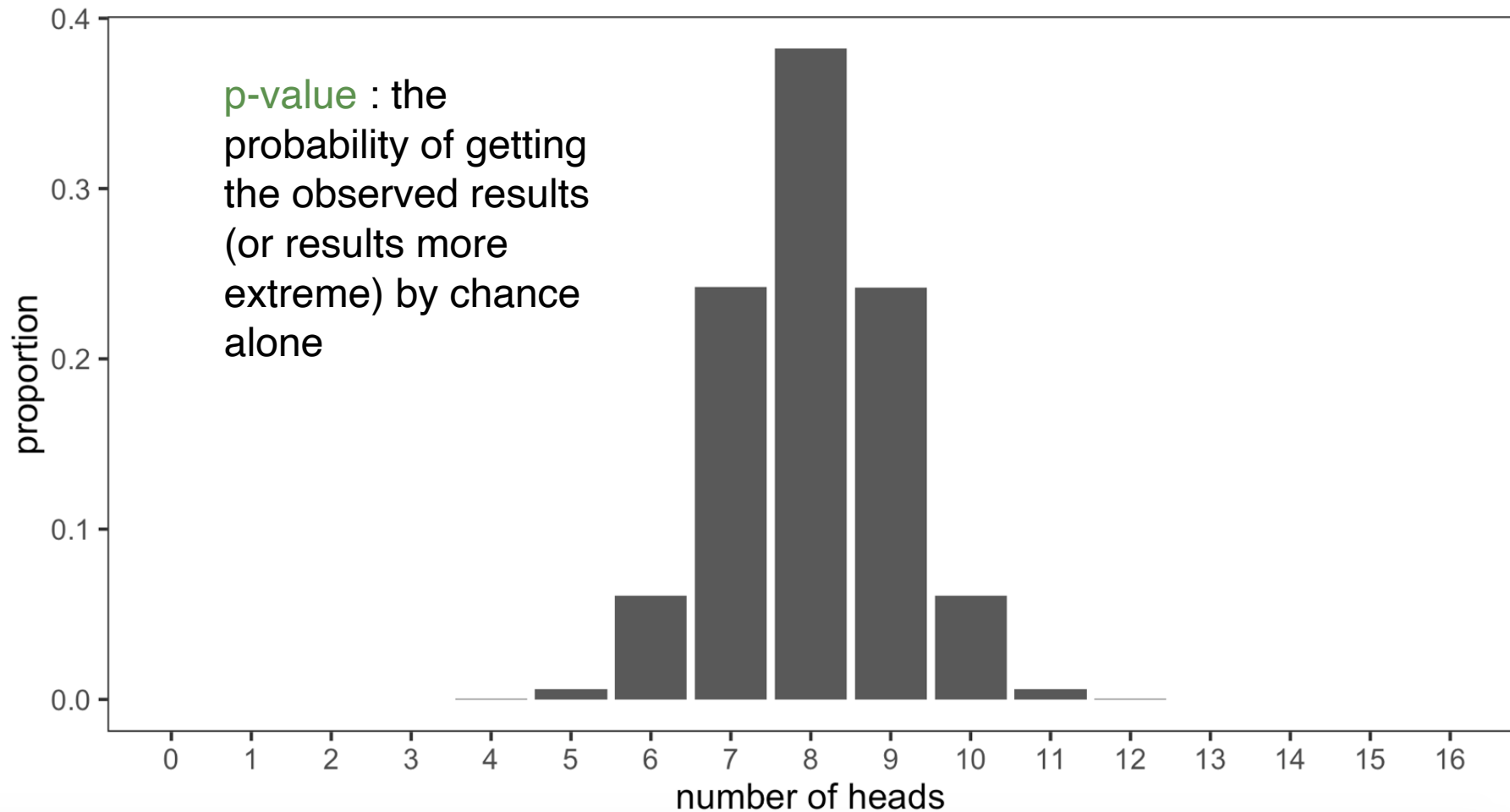
THIS IS NOT TYPE 1 ERROR RATE

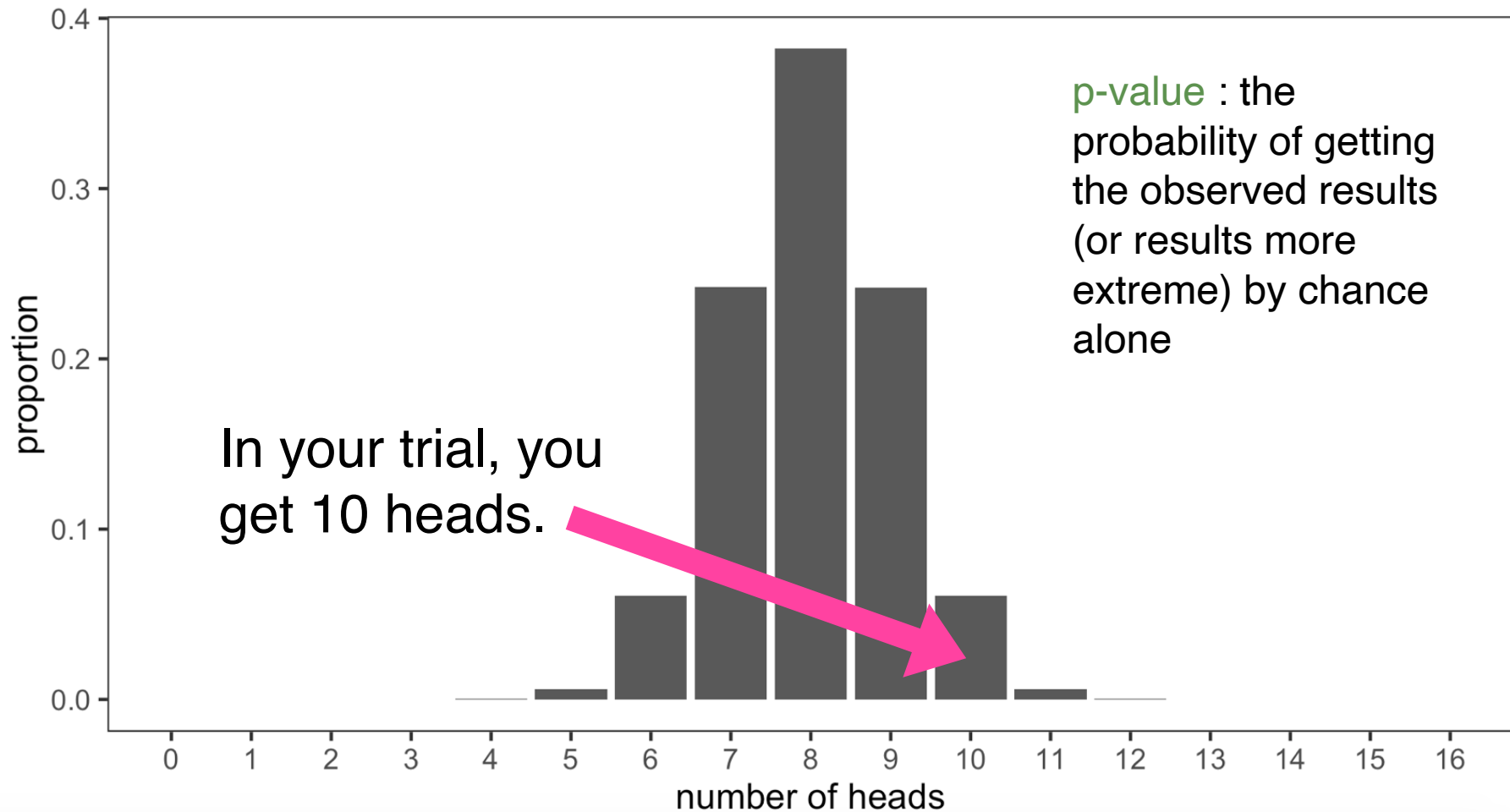
If we flip a coin 16 times and
record the number of heads....
....and then do that 1M times

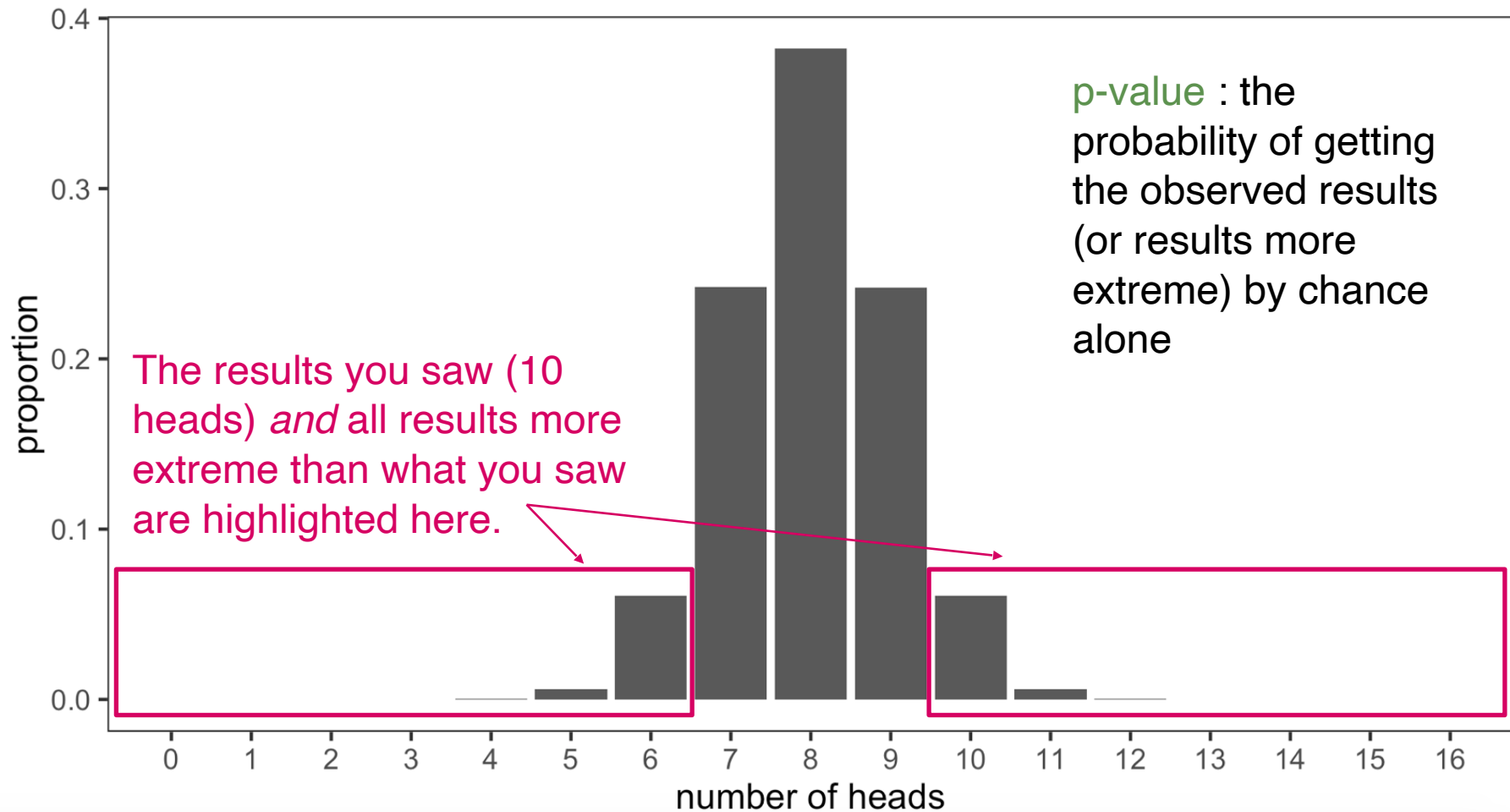
...this is what happens by chance alone.

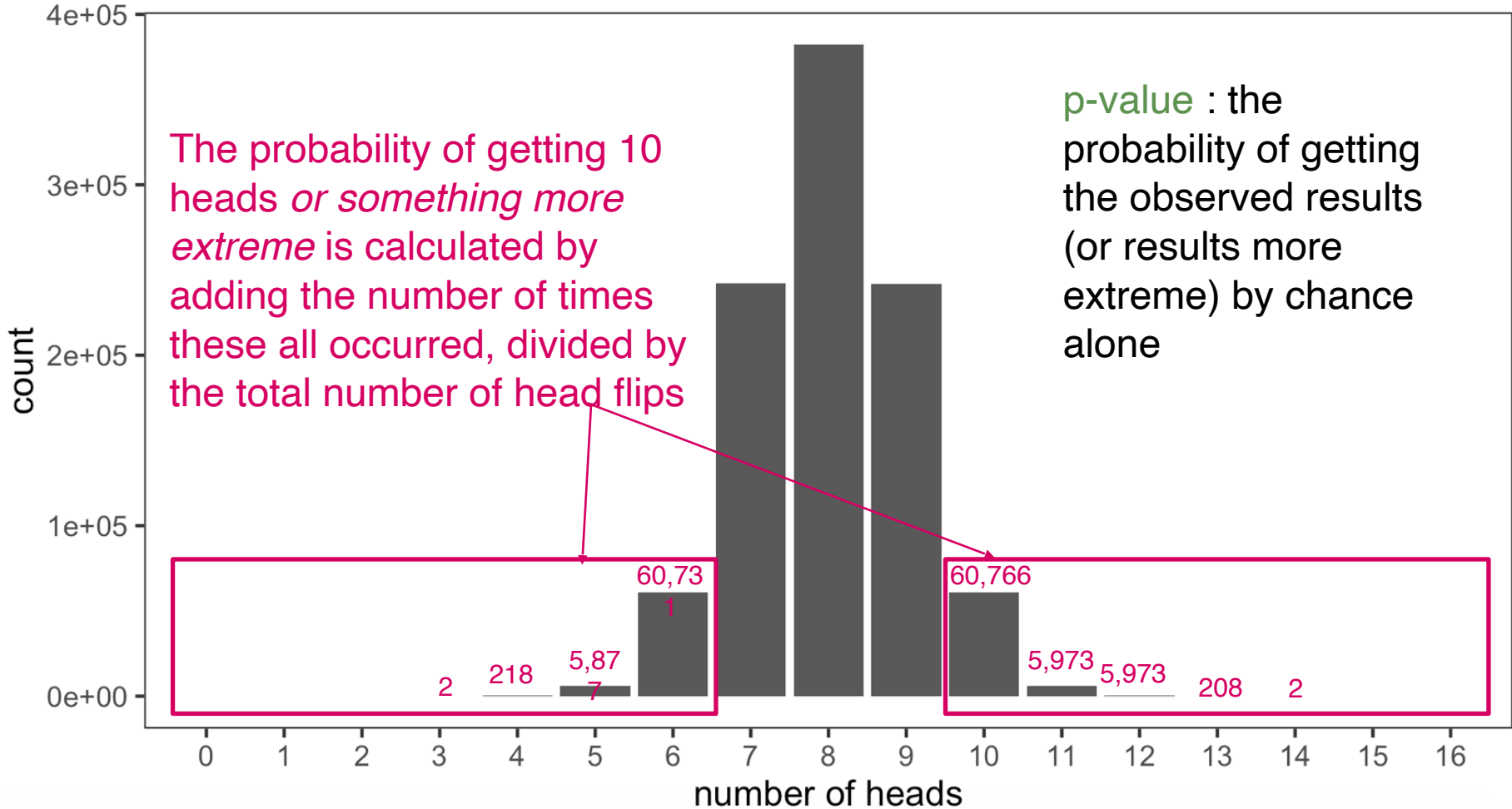


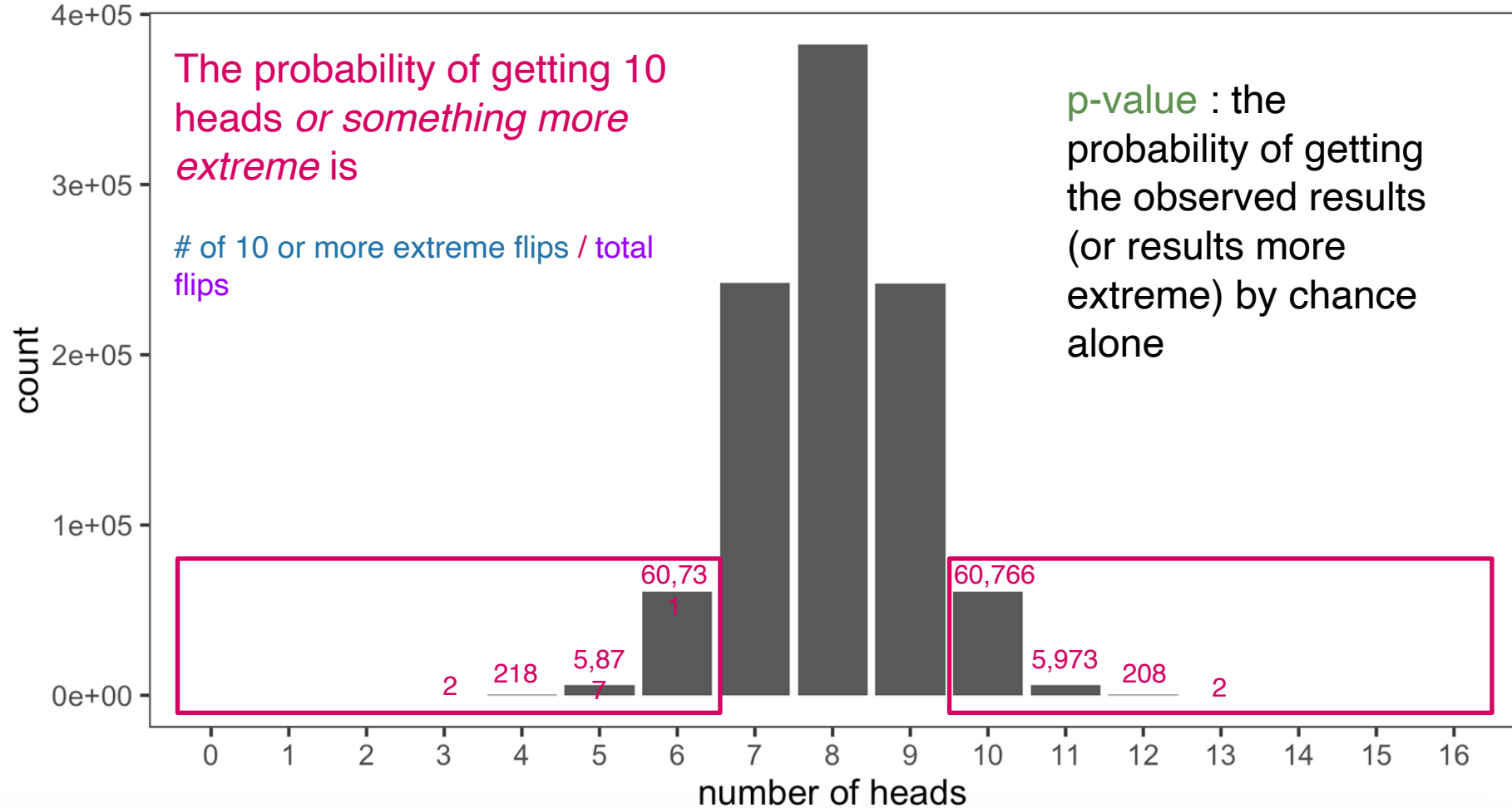


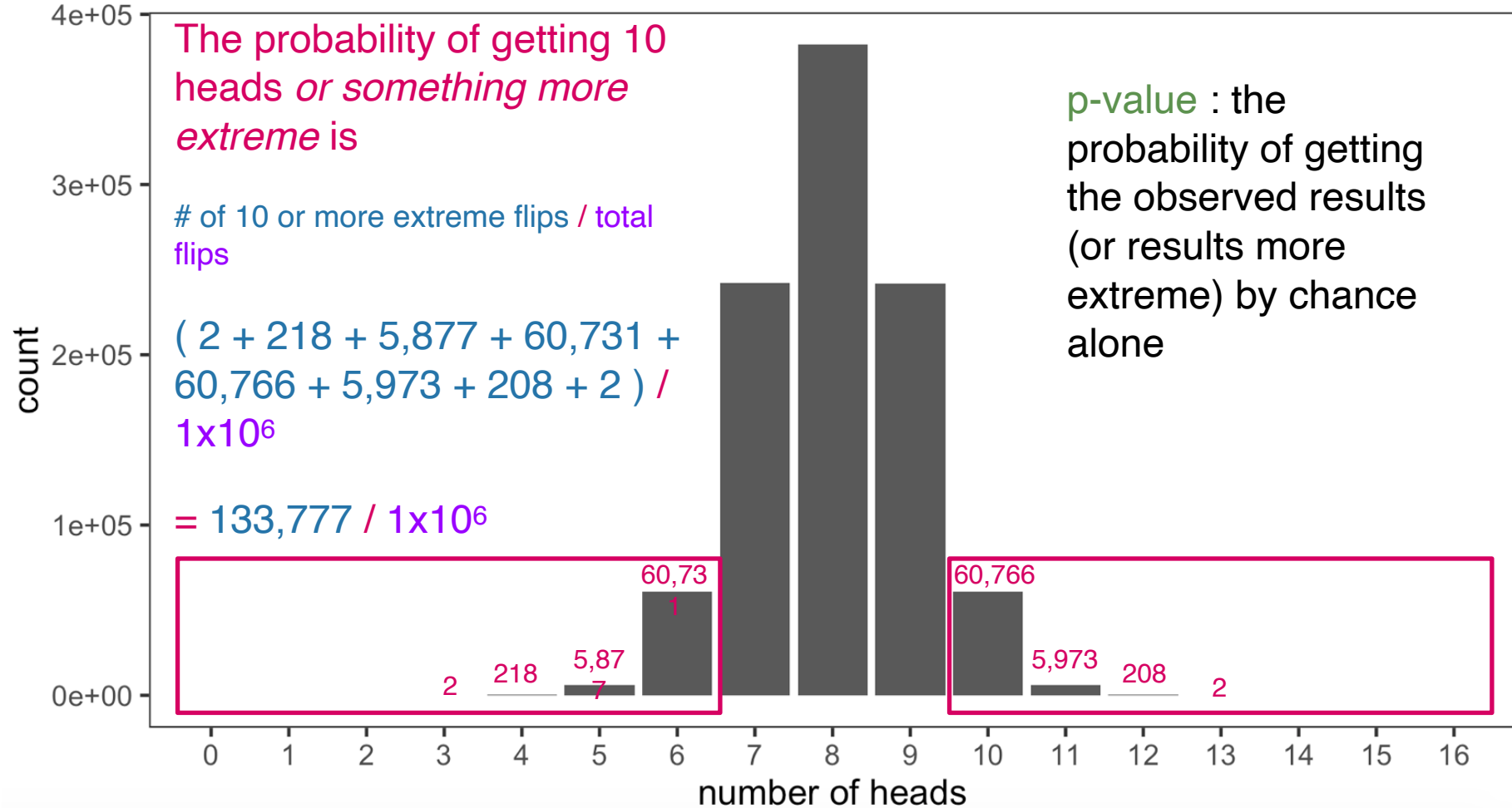


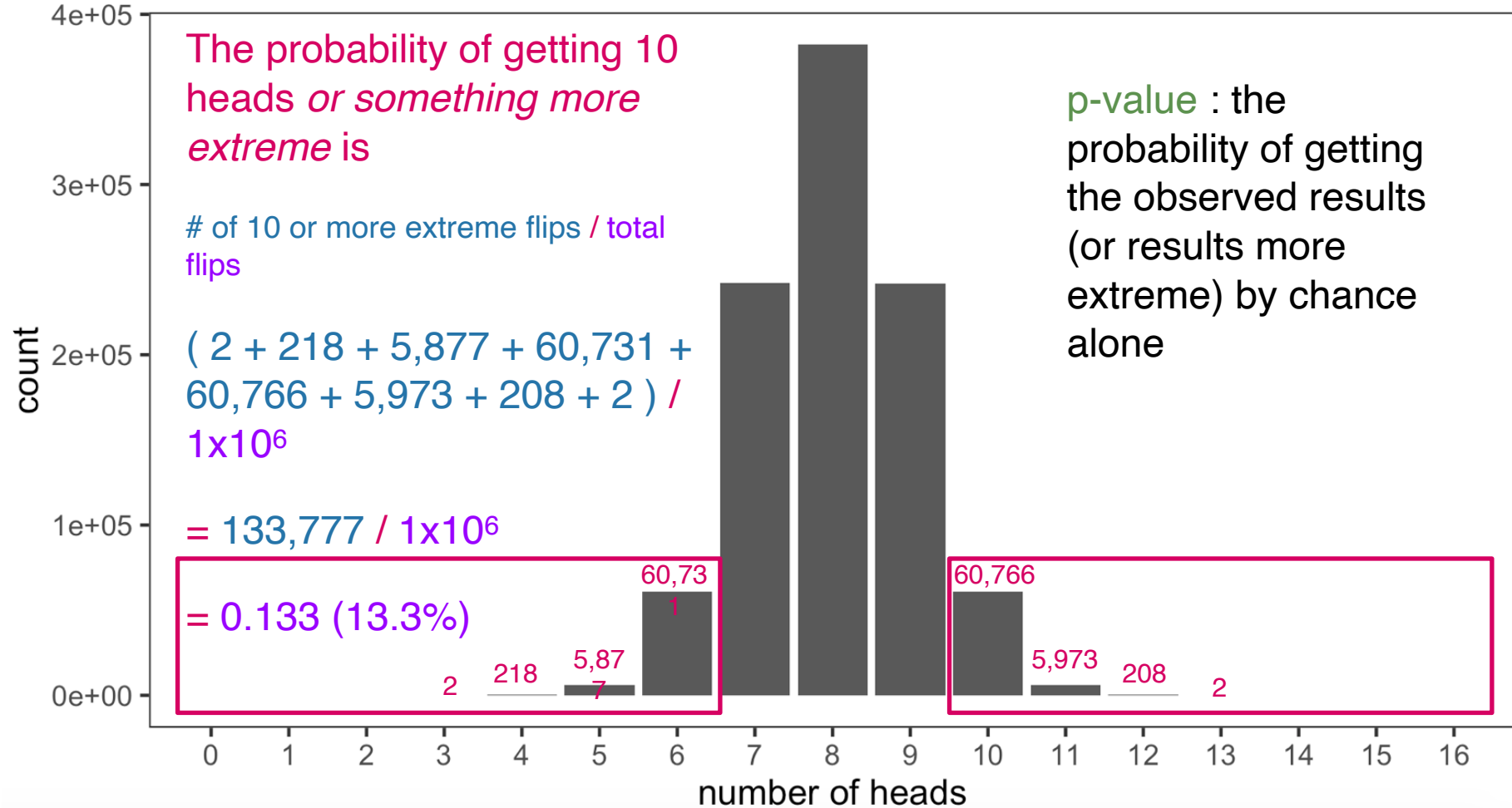


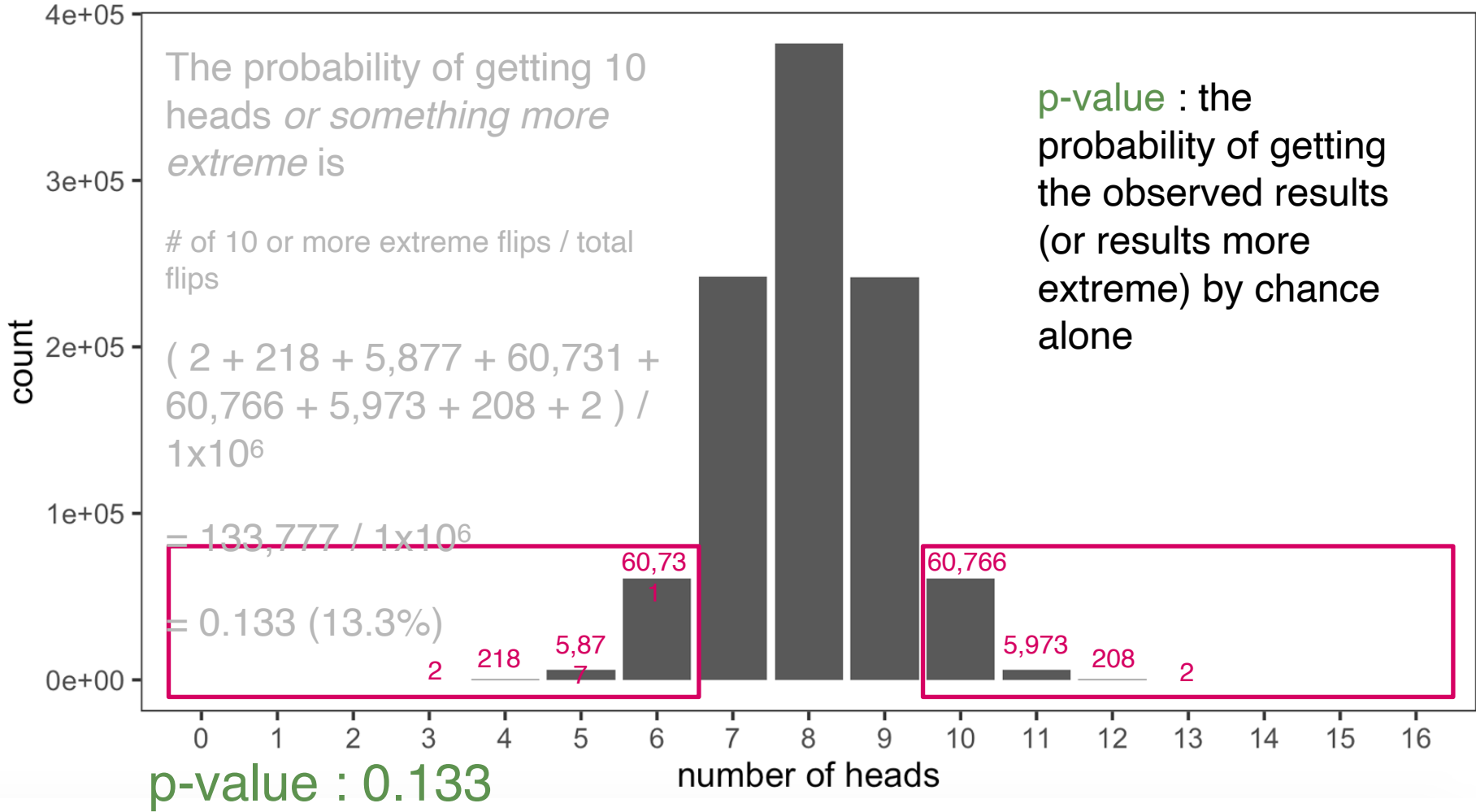


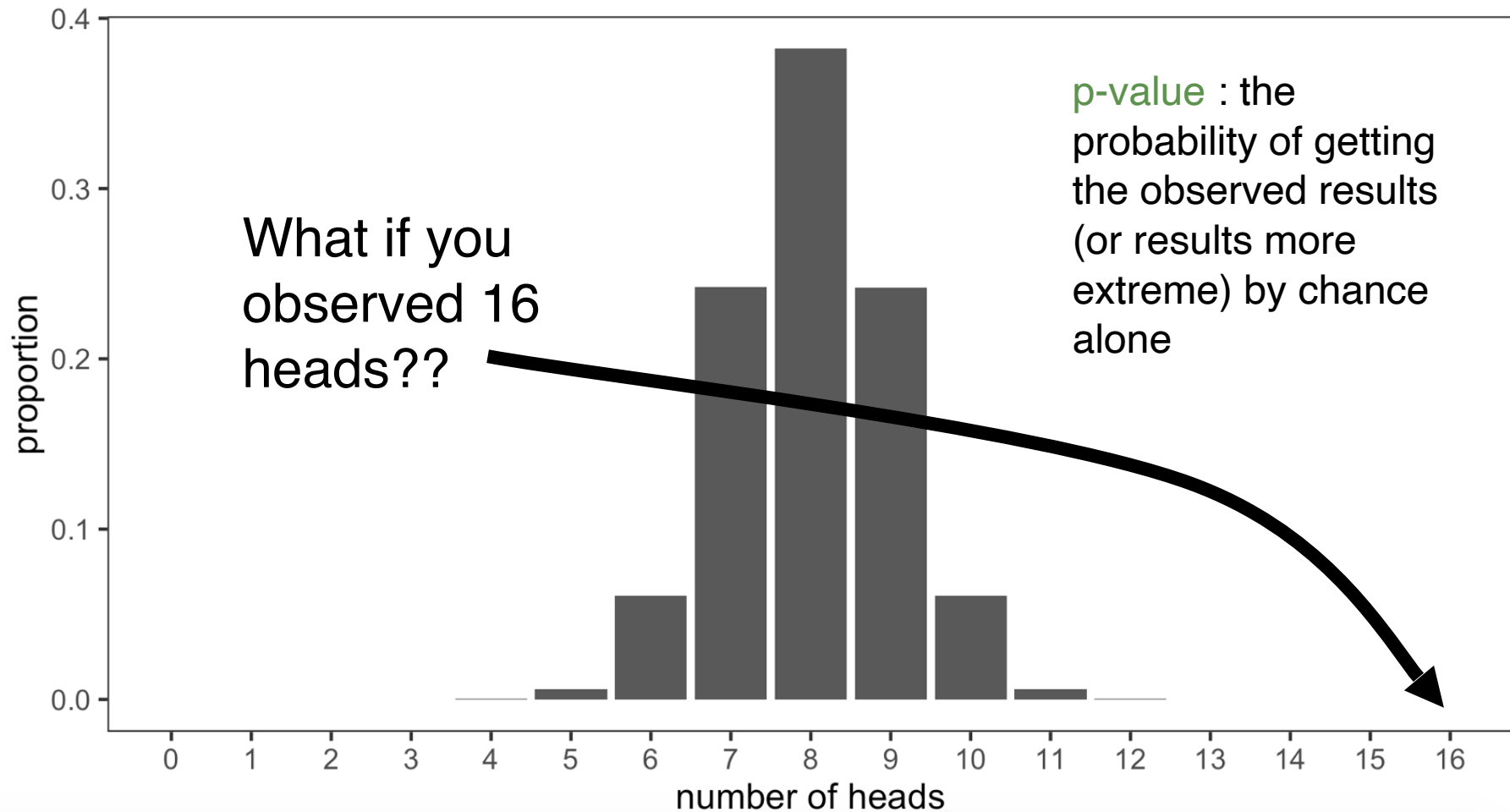


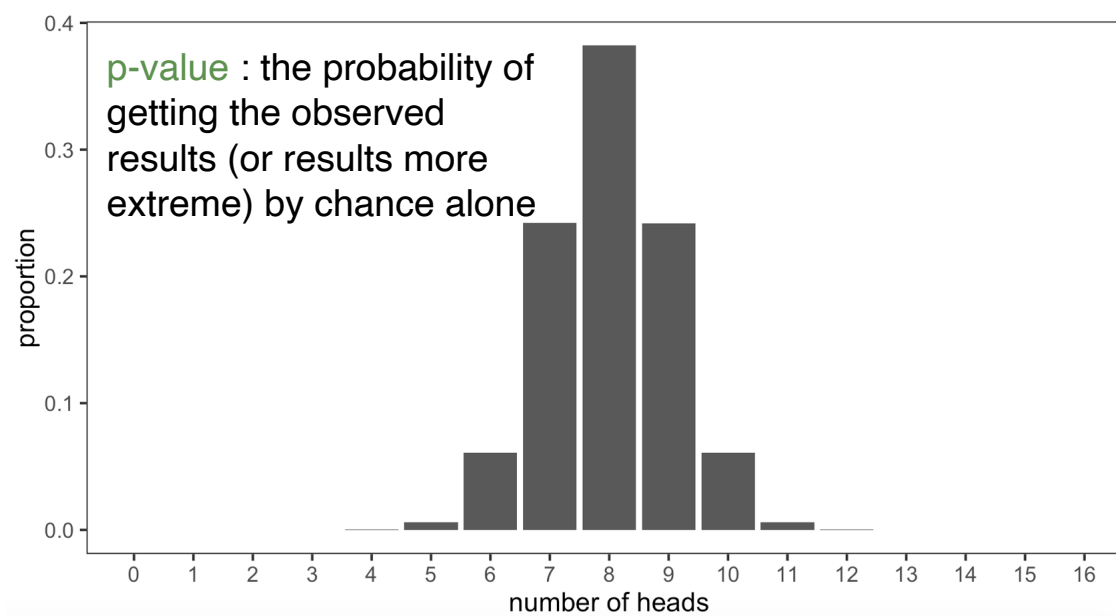












What would be the p-value of you flipping 16 heads?

